A2 Transaction Reconciliation

Level: Medium. Plan for about 6 hours.

The customer

PT Nusantara Retail Group runs a chain of stores in Indonesia. Every month its accounts payable team reconciles supplier invoices against what was booked in the ERP. Three people spend two days on it, and mistakes lead to overpayments. They want a system that does the matching, explains every difference, and leaves the final decision to a person.

What you get

- **Supplier_Invoices_PT_Prima_Kertas_Sep2026.pdf**: the September batch from one supplier, as received.
- **ERP_AP_Export_Sep2026.xlsx**: the company's records, exported from its ERP (read all sheets).
- The business rules below. Treat them as the customer's agreed specification.

Both files are realistic. Expect formatting differences in identifiers, dates and numbers, and expect that either side can contain errors. In production the customer has more than 40 suppliers, each with its own invoice layout. Your sample comes from one supplier.

Business rules

#	Rule
R1	Line amount = quantity x unit price x (1 - discount % / 100), rounded half-up to whole rupiah.
R2	Subtotal = sum of line amounts.
R3	DPP Nilai Lain (other tax base) = subtotal x 11/12, rounded half-up to whole rupiah.
R4	PPN (VAT) = DPP x 12%, rounded half-up to whole rupiah. Total = subtotal + PPN.
R5	A difference of up to IDR 10 on the total, with no line-level differences, is rounding and counts as matched within tolerance.
R6	Credit notes reduce the amount owed and are reconciled like invoices.
R7	Never correct a figure automatically. If a document's own numbers do not add up, or key references disagree, flag it for review.
R8	Use one status per document: MATCHED, MATCHED_WITHIN_TOLERANCE, AMOUNT_MISMATCH, MISSING_IN_ERP, MISSING_IN_PDF, DUPLICATE_IN_ERP, DUPLICATE_IN_PDF or NEEDS_REVIEW. Add reason codes of your own design.

The tax rules are simplified for this exercise. They follow the Indonesian approach in force since 2025 (12% applied to an 11/12 base, an effective 11%) but are not tax advice.

What to build (must have)

1. Extraction of every document in the PDF into structured records, including pages that are hard to read.
2. Normalisation and matching between PDF documents and ERP entries, at document and line level.
3. All arithmetic and matching done in **deterministic code**. Use an LLM only where it adds value, and explain where that is.

4. A **reconciliation report** (CSV or Excel): one row per document with PDF total, ERP total, difference, status, reason codes, a plain-language explanation and a suggested action.
5. A **line-level discrepancy list** and **batch control totals** showing that the overall difference between the two files is fully explained.
6. An **audit trail**: every figure traceable to a PDF page or an ERP sheet and row.
7. Tests for your matching and calculation logic.
8. It must run on a **new batch without code changes**. In the demo round we will give you the October files.

Nice to have

- A review queue or simple UI for the AP team
- Confidence scores and how you would use them
- How you would onboard a new supplier layout safely
- Cost and runtime per batch

What we look for

Area	Strong submissions
Extraction	Reliable structured records from every page, with checks that catch extraction errors.
Judgment	Knows which parts need an LLM and which need plain code. Separates a real discrepancy from formatting or rounding.
Correctness	Right status and cause for each document; totals reconcile across the whole batch.
Auditability	A finance reviewer can verify any result in under a minute.
Communication	Explanations a finance clerk understands; a customer summary an AP manager would forward to their CFO.

Take-home assignment: start here

Thank you for your time in the first conversation. The next step is a take-home assignment. There are three to choose from, and you do **one**. Each is modelled on a real kind of project our engineers deliver for customers in Southeast Asia. All companies, people and data in them are fictional.

The three options

	Assignment	Level	What it emphasises
A1	Document Intelligence: a question-answering assistant over an annual report	Easy	Sound RAG fundamentals, evaluation, honest answers
A2	Transaction Reconciliation: supplier invoices against an ERP export	Medium	Data integrity, deterministic logic, auditability
A3	Product Discovery: a sales assistant over a product catalog website	Hard	End-to-end design: data, search, agents, pricing

Pick the one where you can do your best work. A harder assignment earns no extra points; an excellent A1 beats an average A3. Harder assignments only give more room to show advanced skills.

Time

- Plan for about **6 hours** of focused work. Please do not spend much more; good judgment about scope is part of what we look at.
- You have **5 calendar days** from the day you receive this. The exact deadline (18:00 your local time) is in our email.
- If you need more time for any reason, including a holiday, tell us. Asking is fine.

What to submit

1. A Git repository link (GitHub or GitLab; private is fine if you give us access). Never commit API keys.
2. A **README** with: how to run it in 5 commands or fewer, an architecture overview, key decisions and trade-offs, known limitations, and what you would do with another week.
3. Your **evaluation results** as described in your brief, including cases where your system fails.
4. A **one-page customer summary** (PDF or Markdown) written for the customer's business owner, not an engineer: what you built, how well it works, what it cannot do yet, and what you recommend next.
5. One short README paragraph on **how you used AI tools** while building.

Submit by replying to the email we sent you, with your repository link and your one-page summary.

Rules

- Any language, framework, model provider or database. Free tiers and local models are fine.
- AI coding assistants are allowed. You must be able to explain and change any part of your code in the interview.
- If model API costs are a barrier, tell us and we will give you a capped API key.
- Use the data only for this assignment.
- Questions are welcome: reply to our email and we answer within one business day. Asking good clarifying questions is part of the job.

What happens after you submit

Round	Length	What happens
Demo and deep dive	60 min	You demo your work. We run it on inputs you have not seen. We discuss design, testing, failures and production readiness.
Customer simulation	40 min	A role-played discovery call, a short solution pitch and a conversation with a sceptical stakeholder.
Live engineering	45 min	A small debugging and diagnosis exercise. Bring your usual tools.

Keep your solution runnable for the demo round: we will ask you to process a new input file live, without changing code first.

How we evaluate

Every candidate is scored on the same seven areas, whichever assignment they pick: technical correctness, problem solving and judgment, engineering quality, AI and data evaluation, communication, business and pre-sales thinking, and ownership. We care more about evidence that your system works, and honesty about where it does not, than about polish.